TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	62.9 ± 3.08
Correlation Stability	89.1 ± 0.70
Distribution Consistency	74.0 ± 5.51
Entropy Preservation	68.2 ± 1.17
Mean Z Score Drift	0.5 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	4.3 ± 2.33
Data Density	100.0 ± 0.00
Coverage Score	88.3 ± 1.79
Feature Accuracy	55.0 ± 2.05
Novelty Score	0.6 ± 0.02

TabuLLM: AI Benchmark Report

Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	64.679	+3.845	6.3%	OK
anaemia	0.431	0.270	-0.161	37.4%	Drift
creatinine_phosphokinase	581.839	517.324	-64.515	11.1%	Warn
diabetes	0.418	0.207	-0.211	50.4%	Drift
ejection_fraction	38.084	34.009	-4.075	10.7%	Warn
high_blood_pressure	0.351	0.180	-0.171	48.7%	Drift
platelets	263358.029	257974.797	-5383.232	2.0%	OK
serum_creatinine	1.394	1.779	+0.385	27.6%	Drift
serum_sodium	136.625	134.820	-1.806	1.3%	OK
sex	0.649	0.378	-0.270	41.7%	Drift
smoking	0.321	0.099	-0.222	69.1%	Drift
time	130.261	68.144	-62.117	47.7%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	62.9 ± 3.08
Correlation Stability (*2)	89.1 ± 0.70
Distribution Consistency (*3)	74.0 ± 5.51
Entropy Preservation (*4)	68.2 ± 1.17
Mean Z Score Drift (*5)	0.5 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	4.3 ± 2.33

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	88.3 ± 1.79
Feature Accuracy (*10)	55.0 ± 2.05
Novelty Score (*11)	0.6 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8316666666666667	N/A	0.835	0.00333333333333334	
F1	0.7556715392060958	N/A	0.7684779770485167	0.01280643784242088	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	N/A	0.8533333333333333	0.01833333333333332	
F1	0.7547418227770806	N/A	0.7912750243181139	0.0365332015410333	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7833333333333333	N/A	0.765	-0.01833333333333332	
F1	0.676792588843227	N/A	0.6619160033865916	-0.01487658545663534	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	N/A	0.78	0.02999999999999992	
F1	0.5667171896628354	N/A	0.7014679636738460	0.1347507740110107	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	N/A	0.525	-0.10833333333333332	*
F1	N/A	N/A	0.3187332744333759 6	0.31873327443337596	*

Model: AB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	N/A	0.8366666666666667	0.01166666666666636	
F1	0.7417404771442427	N/A	0.7718718218945454	0.030131344750302647	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8366666666666667	N/A	0.8316666666666667	-0.005000000000000037	
F1	0.7489978613316209	N/A	0.7638515342891115	0.014853672957490538	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	N/A	0.7666666666666667	-0.04	
F1	0.6937863401621888	N/A	0.6863060587344786	-0.007480281427710134	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4599999999999996	N/A	0.4983333333333335	0.03833333333333333	
F1	0.2458200787469080 3	N/A	0.2454181539707855 4	-0.000401924776122464 27	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	N/A	0.515	-0.09833333333333334	*
F1	0.007142857142857143	N/A	0.4348748764690356	0.4277320193261785	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	45.8459018743190 6	0.0	571.0	0.0	14.0	0.0	165589.133375330 74	0.82658548747859 92	127.0	1.0
2	70.3739333308936 6	0.0	685.0	1.0	48.0	0.0	192962.606666910 64	1.20373933330893 68	138.0	0.0
3	84.3713403451041 2	0.0	582.0	0.0	38.0	1.0	193165.690758694 43	1.21257319309791 76	135.0	0.0
4	59.8088382950480 55	0.0	553.0	0.0	37.0	0.0	447431.648174230 36	0.62548822732692 58	138.0	0.0
5	60.0	0.0	429.0	1.0	50.0	0.0	452708.873196829 17	0.88481219947152 94	134.0	1.0
6	74.9745835765039 3	1.0	81.0	0.0	38.0	0.0	367913.584160113 36	3.98627513131212 2	131.0	0.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	77.3356984224580 5	0.0	179.0	1.0	36.0	0.0	198727.917751579 46	1.17879862919299 1	134.0	0.0
2	71.6792151238492	0.0	415.0	0.0	29.0	1.0	618255.507170709 8	1.69168335506275 7	138.0	0.0
3	53.2913689654847 6	0.0	1526.0	0.0	29.0	0.0	312633.095172387 8	1.37338096552243 9	136.0	1.0
4	60.5854641059820 6	0.0	162.0	1.0	30.0	0.0	396579.048619358 8	1.38998155229963 06	136.0	1.0
5	64.0578517684393 5	1.0	320.0	1.0	31.0	0.0	197246.281414751 5	1.34347110610636 07	131.0	0.0
6	70.0	1.0	73.0	0.0	50.0	0.0	351000.0	1.01973272594731 55	134.0	0.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	61.1361731921778 5	0.0	666.0	0.0	25.0	0.0	266655.925774780 9	1.11720371126095 39	134.0	1.0
2	61.3916374067630 4	0.0	568.0	0.0	20.0	1.0	132556.654962705 22	2.91133099254104 35	134.0	1.0
3	71.1916773785658	0.0	202.0	0.0	39.0	0.0	292191.677378565 83	0.95632371844527 89	138.0	1.0
4	65.9277488136308 4	0.0	643.0	1.0	53.0	0.0	193407.225118636 9	1.15927748813630 84	139.0	0.0
5	75.7009590771386 3	1.0	86.0	0.0	37.0	1.0	370803.836308554 6	4.75703580330972 9	131.0	1.0
6	71.8104606836907 8	0.0	111.0	0.0	26.0	0.0	274947.696581546 1	1.02843089744638 38	139.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	79.1000834594393 3	0.0	582.0	0.0	30.0	0.0	263358.03	1.83	134.0	0.0
2	81.4703395201926 4	0.0	750.0	0.0	38.0	0.0	263358.03	1.27889130829010 48	134.0	0.0
3	68.5238350452868 3	0.0	456.0	0.0	33.0	0.0	263358.03	1.73257311298893 06	135.0	0.0
4	70.4615544532443 9	1.0	152.0	0.0	28.0	0.0	279769.170800333 55	0.96153886133110 98	140.0	1.0
5	56.9748040298876 6	1.0	188.0	0.0	36.0	1.0	281037.35846603	1.02250217744569 28	138.0	1.0
6	86.7215479684131 6	1.0	150.0	0.0	48.0	0.0	225243.434146556 9	1.20175089425149 8	136.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	90.84926775706052	0.0	582.0	0.0	38.0	0.0	263358.03	1.83	134.0	0.0
2	88.85386799784278	0.0	582.0	0.0	38.0	0.0	263358.03	1.83	134.0	0.0
3	60.54295894408186	0.0	192.0	1.0	31.0	0.0	400530.0531738005	1.3326282603803148	136.0	1.0
4	51.33476333404777	0.0	793.0	0.0	22.0	1.0	319141.5008081241	1.0929249595937947	136.0	0.0
5	73.10104003855041	0.0	347.0	0.0	46.0	0.0	310296.8466527457	1.166947333440418	133.0	1.0
6	70.91499671509533	1.0	114.0	0.0	32.0	0.0	388469.9220651169	5.687615685576637	134.0	1.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	73.44703915878421	0.0	582.0	0.0	26.0	1.0	263358.03	1.83	134.0	0.0
2	77.38467841475348	0.0	128.0	0.0	48.0	0.0	302573.1052866390707	1.3523064317049305	133.0	0.0
3	60.06127346492863	0.0	451.0	0.0	44.0	0.0	308850.1401754367474	1.6987745307014273	134.0	1.0
4	60.72950506559589	0.0	1490.0	0.0	35.0	0.0	344224.1782926373734	1.8471514906158049	136.0	1.0
5	47.94291504130820404	0.0	540.0	0.0	35.0	0.0	383160.6780991824	1.1471457520654103	144.0	1.0
6	75.79902789532076	0.0	76.0	0.0	38.0	0.0	367360.7776837434	3.92009721046792	131.0	1.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	76.4605884890457 9	1.0	462.0	0.0	40.0	0.0	223539.411510954 2	1.80809117266431 32	142.0	1.0
2	56.9296778424827 54	0.0	1916.0	1.0	28.0	0.0	389238.547402662 75	1.38072629866862 68	136.0	1.0
3	61.3655413250270 2	1.0	156.0	0.0	30.0	0.0	402582.255985573 2	1.11488266644807 95	139.0	0.0
4	64.3083574349313 5	0.0	177.0	1.0	26.0	0.0	490636.888401368 46	1.65985592899311 38	135.0	1.0
5	54.9883860242542 1	0.0	829.0	0.0	21.0	1.0	323946.927585253 5	1.25621876585010 98	136.0	1.0
6	57.9641420870454 24	0.0	337.0	0.0	34.0	0.0	327149.220079132 2	2.64835181503511 84	140.0	0.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	74.5155420986865	0.0	582.0	0.0	20.0	0.0	265080.742983552 3	1.84347991151342 42	130.0	1.0
2	46.5361595656474 6	1.0	225.0	0.0	17.0	0.0	212243.973405725 4	1.40080354590327 68	135.0	0.0
3	81.9042043357030 7	0.0	734.0	0.0	38.0	0.0	263358.03	1.33167819417720 78	134.0	0.0
4	71.2497984906364 5	0.0	155.0	0.0	25.0	0.0	273892.828355805 2	0.99642761186017 36	139.0	1.0
5	52.2696164702474 5	0.0	217.0	0.0	24.0	0.0	211588.731529173 2	2.30853096464454 1	136.0	1.0
6	72.9004299633817	0.0	2808.0	0.0	38.0	0.0	263358.03	1.1	134.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	74.9364216052095	0.0	1508.0	0.0	42.0	0.0	236807.0286801289	2.16118212247583	134.0	0.0
2	61.5622808863913	0.0	432.0	0.0	45.0	0.0	308678.5964701267	1.66875438227217	134.0	1.0
3	74.4470944014930	0.0	582.0	0.0	30.0	0.0	225276.4527992535	1.98389205825110	134.0	1.0
4	75.0772854267401	0.0	430.0	1.0	39.0	0.0	227421.2109117390	2.77651363967379	134.0	1.0
5	60.8927316860124	0.0	139.0	1.0	34.0	1.0	279066.8963299234	1.60357561046625	131.0	0.0
6	68.9117467394472	0.0	124.0	0.0	44.0	1.0	283564.6986957789	1.28911746739447	136.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	63.2741084523400	0.0	134.0	1.0	37.0	1.0	280356.8087450175	1.52419638492199	132.0	0.0
2	64.2788389922088	0.0	2516.0	0.0	44.0	0.0	263358.03	4.67341702087217	140.0	0.0
3	50.5573606117785	1.0	193.0	0.0	20.0	0.0	216209.0102294169	1.24651649617638	136.0	0.0
4	59.7102476292088	0.0	518.0	0.0	37.0	0.0	217289.7523707911	3.33780953651108	134.0	1.0
5	84.6877368388324	0.0	128.0	0.0	59.0	0.0	305859.4815774745	1.20624526322335	132.0	0.0
6	67.3886378248635	0.0	125.0	1.0	37.0	1.0	290789.5416553548	1.45668173050818	136.0	0.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0